WINNERS ONLINE SCHOOL

CHEMISTRY PRACTICE QUESTIONS

BY

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FAILING OR PASSING EXAMS IS A CHOICE. MAKE A DECISION TO PASS BY JOINING WINNERS OLINE SCHOOL TO HIT YOUR TARGET

1.The Table below shows electron configurations of elements R, S, T, U, V, W, X, Y and Z

LEMENT	**ELECTRON CONFIGURATION**
R	2,7
S	2,8,6
T	2,8,2
U	2,4
V	2
W	2,8,7
x	2,1
y	2,6
z	2,8,8

(a) Which elements in the Table belong to period 2 of the periodic table?

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(b) Give a reason for the answer in

(c) State three ways in which atoms attain stability

(d) What type of bonding would exist when element R reacts with element T '?

(e) Explain how this bonding occurs

(f) Write down the chemical equation for the reaction that would occur between R and T.

(g) Define atomic radius

(h) In terms of atomic radius, explain the difference in reactivity between **R** and **W.**

(i) Explain why element V, which has 2 valance electrons, is taken as a group 8 element

(j) What is the formula of a compound that is formed between **X** and **Y**?

(k) How can element **S** attain a stable configuration?

(l) Which any two elements can form positive ions?

(m) Give a reason for the answer

(n) Give any three properties of element **X.**

(o) What would happen if element **X** was mixed with water?

(p) Give a reason for the answer

(q) Give any letter of an element in the table that would not react with another element.

(r) Give a reason for the answer

2. A certain element could be represented as $^{23}_{11}X$

(i) To which group does **X** belong? Give a reason.

(ii) Identify element **X** in Periodic Table

(iii) Write the electron configuration of this element

3. Why are metals good conductors of heat?

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4. State two differences between ionic compounds and covalent compounds.

5 (i). Define the term "allotropes

(ii) State two allotropes of sulfur.

5. Table2 shows the atomic numbers, melting points, boiling points and atomic radius of some elements.

LEMENT	**atomic numbers**	**melting points**	**boiling points**	**atomic radius**
R	17	-105	-44	0.078
S	9	-200	-188	0.062
T	53	123	192	0.133
U	35	-8	72	0.112

(i) Which elements are liquid and solid at 25°c

(ii) Why does T has the biggest radius?

(iii) Mention nay two chemical properties of halogens.

6 (i) Define "**concentration"** of a solution.

(ii) 300mg tablet of a drug was completely dissolved in 10ml of water. The molecular formula of the drug is $C_9H_8O_4$.

(iii) Calculate the number of moles in the tablet (RAM: C=12, H=1 and O=16)

(iv) Calculate the concentration of the solution in moles per liter.

7.Calculate the volume of 0.1M sodium hydroxide that is needed to neutralize 20 cm^3 of 0.1M hydrochloric acid.

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8. In a titration, 20 $cm^3$of hydrogen chloride solution reacted completely with 25 cm^3

of 0.2M sodium hydroxide solution mixed with phenolphthalein solution.

(i) Explain the function of phenolphthalein in the titration.

(ii) Which was the standard solution in the titration? Give a reason.

(iii) Write a balanced equation for the reaction between sodium hydroxide and hydrochloric acid.

(iv) Calculate the concentration of the acid.

9. Describe how the concentration of 20$cm^3$of sodium chloride solution can be determined by

evaporation method. State two sources of error

10.What is **a "mole"?**

(ii) A solution was made by dissolving 8g of sodium hydroxide in 100cm^3

of water. Calculate the molarity of the solution. (RAM: Na=23, O=16, H=1)

11.What is meant by "empirical formula of a compound"?

(ii) Work out the empirical formula of a compound that has the following percentage composition by mass of elements: C═40%, H═6.67%, and O═53.33%. (RAM C=12, H=1, 0=16) (

12. Describe how **250cm^3**of a **1M** copper sulphate solution could be prepared using hydrated copper sulphate crystals ($CuSO_4$ - $5H_2O$)

13. Define **"molar volume" of a gas**.

(ii) Calculate the number of moles of Sulphur dioxide occupying 120dm^3at room temperature and pressure. (Molar volume at room temperature and pressure is 24 dm^3)

14.Explain why bond breaking is endothermic while bond making is exothermic.

(i) Draw an energy level diagram for the following chemical reaction:

$NaOH_{(aq)} + HCl_{(aq)} + NaCl_{(aq)} + H_2O_{(l)}$, H= - 57 kJ/mol

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(ii) Is the reaction in endothermic or exothermic?

(iii) Give a reason for the answer ?

(iv) What is the meaning of (aq) and (l) in the equations?

15.Define **"electroplating"**

Iron(Fe) displaces copper(Cu) from copper sulphate solution($CuSO_4$).

(i) Write down a balanced chemical equation for the reaction.

(ii) What is the reducing agent in the reaction?

(iii) Give a reason for the answer

16. What is the difference between "oxidation" and "reduction"?

(ii) Mention any two ways of preventing corrosion in metals.

(iii) Calculate the oxidation number of nitrogen (N) in a nitrate ion NO_3^- given that the oxidation number of oxygen (O) is 2.

17.The chemical equation below shows displacement reaction between zinc (Zn) and lead nitrate $Pb(NO_3)_2$.

Zn + Pb(NO3)2 ⟶ Zn(NO3)2 + Pb.

Name the reducing and oxidizing agents in the reaction.

FIGURE below shows an experiment that Edina a student of **Winners school** conducted in the lab. Study very carefully and answer the questions that follow

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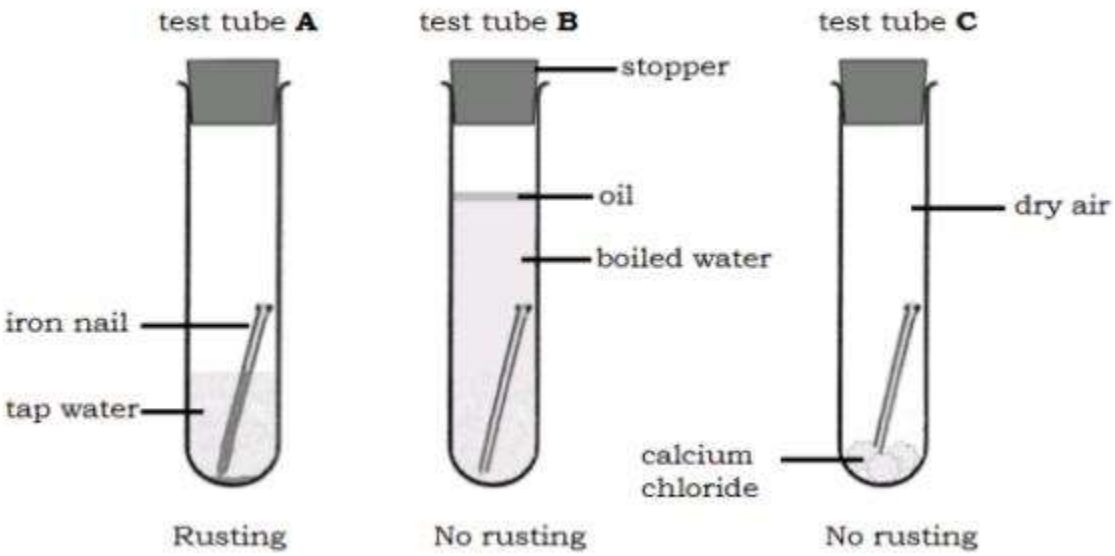

(i) Why was anhydrous calcium chloride used in tube **C**?

(ii) Why was water in tube **B** boiled?

(iii) State any two conditions that could be the possible cause for the results in tube A.

(iv) Explain any four points how this result in tube a can be avoided

18.With the aid of a labelled diagram, explain how a silver spoon could be electroplated using copper. In the explanation, include the half equation for the reaction at the cathode.

19.Define an "acid" according to Lowry/Bronsted theory.

(ii) Describe how a hydronium ion (H_3O+) is formed.

(iii) Water reacts with the carbonate ion according to the following chemical equation.

$H_2O\ (l) + CO_3^{2-} \leftrightharpoons OH^- + H_2CO_3-$

(i) Give one conjugate acid-base pair in the reaction.

(ii). Ammonia is an example of a strong base.

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(iii) What is a “strong base”?

(iv) Write a chemical equation to show the ionization of ammonia in water.

(v) Identify one conjugate acid-base pair from the equation

20 (i) differentiate between soft and hard water

(ii). Mention any three disadvantages of water hardness

(iii) With the well labelled diagram discuss ion exchange as the method of removing water hardness

21. (i) Define excess reagent

(ii). Sodium reacts with chlorine to form sodium chloride**. Write** the balanced chemical equation for this reaction

(iii) if 8g of sodium reacts with 2g of chlorine, determine limiting reagent

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